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HBV A₁₇₆₂T, G₁₇₆₄A mutations are a valuable biomarker for identifying a subset of male HBsAg carriers at extremely high risk of hepatocellular carcinoma: A prospective study

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Abstract

OBJECTIVES: Surveillance of hepatocellular carcinoma (HCC) can detect small tumors for resection but at a huge cost of health resources. The challenge is to reduce the surveillance population. We reported that 96% of HCC patients but only 24% of controls were infected with the hepatitis B virus (HBV) with A₁₇₆₂T, G₁₇₆₄A mutations in Guangxi, China. It is likely to be extremely beneficial in terms of cost and resources if a significant number of tumours can be detected early by screening this selected population. Our aim is to test this hypothesis.

METHODS: A cohort of 2258 hepatitis B surface antigen positive subjects aged 30-55 was recruited in Guangxi. Following evaluation of virological parameters at baseline, HCC is diagnosed by six-monthly measurements of serum alpha fetoprotein levels and ultrasonographic examinations.

RESULTS: 61 cases of HCC were diagnosed after 36 months of follow-up. The HCC rate was higher in the mutant than wild type group ($p < 0.001$, rate ratio (RR) 6.23 [95% confidence interval (CI) 2.83, 13.68]). The HCC rate in the male mutant group was higher than that in the male wild type group ($p < 0.001$, RR 11.54 [95% CI 3.58, 37.24]). Specifically, 93.3% of male cases are infected with the mutant. Multivariate analyses showed that in males, increasing age and A₁₇₆₂T, G₁₇₆₄A double mutations are independently associated with developing HCC.

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Specific author contributions: Zhong-Liao Fang: Carried out serological assays and PCR and sequencing for HBV DNA. Participated in the design of the study, carried out the statistical analysis, drafted and revised the manuscript critically. Caroline A Sabin carried out the statistical analysis. Bai-Qing Dong, Shao-Chao Wei and Kong-Xiong Fang organized the field works for the Long An cohort. Lian-Ying Ge evaluated individuals with HCC for surgery. Qin-Yan Chen and Jin-Ye Yang carried out serological assays and PCR for HBV DNA. Xue-Yan Wang assisted in processing the data. Tim J Harrison designed the study, analysed the data, and wrote the manuscript.

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CONCLUSIONS: HBV A₁₇₆₂T, G₁₇₆₄A mutations constitute a valuable biomarker to identify a subset of male HBsAg carriers aged >30 at extremely high risk of HCC in Guangxi, and likely elsewhere.

Keywords

hepatitis B virus; polymerase chain reaction; nucleotide sequences; viral load; mutation

INTRODUCTION AND BACKGROUND

Hepatocellular carcinoma (HCC) is the fourth most common cause of cancer mortality worldwide and the third most common in men (1). The incidence of HCC is not uniform across the world but varies according to the prevalence of underlying liver diseases. The highest incidence of HCC is seen in China (2). Surgery remains the only curative modality for HCC but a large percentage of cases are not suitable for surgery because of intrahepatic or distant metastases at the time of diagnosis. However, five-year survival rates of 61.3% have been achieved for resection of small HCCs in China (3). Recent studies suggest that surveillance of HCC by regular testing for serum alpha-fetoprotein (AFP) and/or ultrasound (US) imaging can detect HCC at an earlier and (probably) resectable stage, which may be translated into improved survival (4, 5). However, a population-based HCC surveillance program would place a huge demand on public health resources and be impractical in most countries. The challenge is how to identify the population at highest risk of HCC. Chronic hepatitis B virus (HBV) infection is the most important etiology of HCC in Asia (6). In China, individuals aged over 35 years and with serological evidence of HBV infection (positive for hepatitis B surface antigen, HBsAg) are currently considered to be the population at highest risk for HCC (5).

The mechanisms of oncogenesis of HBV remain obscure. However, attention has been focussed on viral factors associated with the development of HCC, including HBV genotype, viral load and the occurrence of specific mutations in the HBV genome. HBV can be divided into eight different genotypes (designated by capital letters A to H) based on an intergroup divergence of 8% or more in the complete nucleotide sequence (7, 8). HBV genotypes vary in their geographic distribution and have been reported to have clinical relevance (9). Genotypes B and C are predominant in Asia (10) but their precise significance in terms of development of HCC remains controversial (10-13). Several studies have shown that high viral loads, specifically, titres of viral DNA in serum above 10⁴⁻⁵ genome copies/ml, are associated with the development of HCC (10, 14, 15). The common precore mutation (G₁₈₉₆A), mutations in enhancer II (C₁₆₅₃T) and the basal core promoter (T₁₇₅₃V and the double mutations, A₁₇₆₂T, G₁₇₆₄A), and deletions in the pre-S region (13, 14, 16) have been reported to be associated with the development of HCC. Perhaps the most convincing association is with HBV with double mutations in the basal core promoter (BCP) (17-21).

In our initial study (17), we found BCP double mutations in HBV DNA from serum and/or tumour tissue of 13 of 14 HCC patients but only 3 of 11 asymptomatic controls. In a follow-up study (18), 35 of 36 HCC patients, but only 9 of 39 asymptomatic carriers, were found to be infected with HBV with BCP double mutations. Overall, we found the mutations in HBV from 96% of HCC patients, but only 24% of controls, from Guangxi, China.

As mentioned above, the key to achieve cost-effectiveness for screening is to identify the population at highest risk and currently, in Guangxi, China, it is practical to consider HBsAg positive males aged >30 as the population at highest risk (22-24). Our previous studies suggest that only around one quarter of them are infected with HBV variants with double

mutations in the BCP. (18) Therefore, it is likely to be extremely beneficial in terms of cost and resources if a significant number of tumours can be detected early by screening this selected population.

We hypothesize that, in Southern China and perhaps elsewhere in the world, BCP double mutations constitute a valuable biomarker for screening carriers of HBsAg in order to identify the subset at highest risk of developing HCC. To test this hypothesis, we have embarked on a prospective study of HBsAg-positive adults living in Long An county, Guangxi, one of the regions of China with the highest annual incidence of HCC (25). Here, we report the results of 36 months of follow-up.

STUDY SUBJECTS AND METHODS

Study population

Study subjects were recruited from agricultural workers aged 30-55 living in the rural area of Long An county, Guangxi, China. Informed consent in writing was obtained from each individual. The study protocol conforms to the ethical guidelines of the 1975 Declaration of Helsinki and has been approved by the Guangxi Institutional Review Board and the UCL Committee on the Ethics of Non-NHS Human Research (Project number 0042/001).

Our Chinese study team comprises doctors from the Centres for Disease Prevention and Control (CPDC) of Long An county and the CPDC of Guangxi Province. Individuals aged 30 to 55 years resident in 128 villages throughout Long An county were selected for testing for HBsAg using stratified sampling. In brief, the entire county was divided into 128 strata according to village and 25% of individuals aged 30 to 55 were selected randomly from each stratum. From 1st March, 2004, the study teams travelled to 128 villages in each of the 12 townships of Long An county to visit and inform agricultural workers aged 30-55 and take a 3 ml sample of blood by venepuncture for screening. All samples were tested for HBsAg and those positive were tested in China for AFP to exclude pre-existing HCC and for HBV DNA using nested PCR. We also detected anti-HCV and excluded those positive to eliminate the confounding effect of HCV infection on the incidence of HCC. We started to follow up our study subjects from 1st July, 2004. Each study subject completed a one-page questionnaire at the first visit and provided a serum sample every six months for the assessment of virological parameters and AFP concentrations, and is monitored for HCC by US. All cases of HCC diagnosed were confirmed at the Medical University of Guangxi, the Cancer Institute of Guangxi or the Hospital of Guangxi (Nanning). The diagnosis was made by one or more of the following: (i) surgical biopsy; (ii) elevated serum AFP (levels ≥ 400 ng/ml), excluding pregnancy, genital cancer and other liver diseases including metastasis of tumors from other organs, plus clinical symptoms or one image (US or computed tomography, CT); (iii) elevated serum AFP (levels < 400 ng/ml), excluding pregnancy, genital cancer and other liver diseases including metastasis of tumors from other organs, plus two images (US and CT) or one image (US or CT) and two positive HCC markers such as DCP, GGT II, AFU, CA19-9 etc.

Serological testing

Sera were tested for HBsAg, HBeAg/anti-HBe, anti-hepatitis C virus (HCV) and alpha fetoprotein (AFP) using enzyme immunoassays (Zhong Shan Biological Technology Company, Limited, Guangzhou, China). Alanine aminotransferase (ALT) levels were determined using a Reitman kit (Sichuan Mike Scientific Technology Company, Limited, Chengdu, China).

Nested PCR for HBV DNA and Nucleotide Sequencing

DNA was extracted from 85 μ l serum by pronase digestion followed by phenol/chloroform extraction. First round PCR was carried out in a 50 μ l reaction using primers B935 (nt 1240–1260, 5'-AACCTTTGTGGCTCCTCTG-3') and MDC1 (nt 2304–2324, 5'-TTGATAAGATAGGGGCATTG-3'), with 5 min hot start followed by 30 cycles of 94°C for 30 sec, 50°C for 30 sec, and 72°C for 90 sec. Second round PCR was carried out on 5 μ l of the first round product in a 50 μ l reaction using primers CPRF1 (nt 1678–1695, 5'-CAATGTCAACGACCGACC-3') and CPRR1 (nt 1928–1948, 5'-GAGTAACTCCACAGTAGCTCC-3') and 5 min hot start, then 30 cycles of 94°C for 30 sec, 55°C for 30 sec, and 72°C for 30 sec.

The purpose of the nested PCR above was to derive as much nucleotide sequence data as possible from the BCP and surrounding region. However, the BCP is very close to the 5' end of the minus strand of genomic DNA and the PCR spans that discontinuity, resulting in a detection limit of around 10^3 – 10^5 genomes/ml (and this is confirmed by our viral load data). Therefore, negative samples were retested using a modified nested PCR that did not span the 5' ends of the genome. The primers were (first round) P2 (26) (nt1823–1806, 5'-CCGGAAAGCTTGAGCTCTTCAAAAAGTTGCATGGTGCTGG-3') and XSEQ1 (nt1547–1569, 5'-CCCCGTCTGTGCCTTCTCATCTG-3') and (second round) CPRF1 and MDN5R (nt1794–1774, 5'-ATTTATGCCTACAGCCTCCT-3'); otherwise, the parameters for PCR were as described above. To determine the sequence at nucleotide position 1653, samples were amplified using the original PCR described above but substituting primer XSEQ1 for CPRF1 in the second round.

Products from the second round were confirmed by agarose gel electrophoresis and then purified using the GenElute™ PCR Clean-up Kit (SIGMA, St. Louis MO, USA) according to the manufacturer's instructions. Cycle sequencing was carried out directly on 2 μ l purified amplicons DNA using primer CPRR1 (products from original PCR) or CPRF1 (products from modified PCR) and a BigDye® Terminator V3.1 Cycle Sequencing Kit (Applied Biosystems, Foster City, CA, USA) according to the manufacturer's instructions.

Statistical Methods

Statistical comparisons of the prevalence of BCP double mutations between males and females and across the different age groups and the analysis of the occurrence of nucleotide 1653 and 1753 mutations in individuals with HCC and controls were performed using Chi-squared tests. Associations between pre-existing double mutations and the development of HCC were tested for significance using Poisson regression models, after adjusting for the potential confounding effects of age and sex, and with the (log) follow-up time offset, using SAS version 9.1. For these analyses, follow-up was considered from the time of entry in the study until the development of HCC or last follow-up visit, whichever occurred first. All P-values were two-tailed, and $P < 0.05$ was considered to be significant.

RESULTS

Grouping of study subjects

In order to identify study subjects, 27,340 individuals in Long An county, Guangxi were tested for HBsAg; 14.5% were positive. The overall prevalence was significantly higher in males (16.7%) than females (12.6%) and declined markedly with age in both genders (Table 1). All HBsAg positive serum samples, except those that were AFP positive, were tested in China for HBV DNA using a nested PCR designed to amplify the BCP and precore region. 1596 were positive and included in the study cohort. In order to ensure sufficient samples for stratification analysis, we enrolled according to stratified sampling 1402 HBsAg positive

individuals who were found HBV DNA negative in China. These samples were retested in London using a more sensitive, modified nested PCR (see Methods) to amplify the BCP whilst avoiding the gap in genomic HBV DNA. A sufficient quantity of the baseline sample remained for this analysis in 1,033 of the 1,402 cases. Only 246 samples were negative using both assays. The nucleotide sequences of the products of both sets of positive PCR reaction were determined (the sequence of 29 samples could not be determined and therefore were grouped as HBV DNA not detected): 1001 were wild type at positions 1762/1764, 1257 had the specific mutations A₁₇₆₂T, G₁₇₆₄A, and 96 had other mutations at these positions or were positive for anti-HCV. The last group included 69 with the single mutations A₁₇₆₂T or G₁₇₆₄A, 12 with other single or double mutations at these positions, and 8 with deletions of up to 9 nucleotides (Table 2).

We divided the study subjects into the mutant group (double mutations, 1261 individuals) and wild type group (997 individuals). Their mean ages were 39.5±6.8 years (± standard deviation) and 37.6±7.1. With further stratification according to gender, there were 702 males in the mutant group and 561 in the wild type group, and their mean ages were 38.6±6.6 and 37.1±6.9, respectively. There were 559 females in the mutant group and 436 in the wild type group and their mean ages were 40.6±6.9 and 38.3±7.4, respectively (Fig. 1). Of the 2998 recruited initially, 740 were excluded: 638 because the BCP sequences could not be determined at baseline, 13 who were anti-HCV positive and the 89 whose BCP sequences were neither wild type nor had the specific mutations A₁₇₆₂T or G₁₇₆₄A. As controls, we also followed up 2460 HBsAg-negative individuals matched to the individuals in the study cohort in terms of sex, age and neighborhood.

The epidemiological characteristic of basal core promoter mutations

In order to confirm the accuracy and reproducibility of the tests for the basal core promoter double mutations, we re-amplified and sequenced the BCP region of HBV from 99 study subjects using the same baseline serum samples and the results are consistent. The prevalence of the mutations A₁₇₆₂T, G₁₇₆₄A is the same (53.7%) in males and females (Table 2). The prevalence tends to increase up to 35 years of age (Table 3) and parallels a decline in the prevalence of HBeAg and a rise in anti-HBe (Fig. 2). These presumed seroconversion events seem to be particularly common between the ages of 25 and 35. However, because the BCP mutations down-regulate, rather than abrogate completely, the synthesis of HBeAg, the correlation between anti-HBe and the BCP mutations is somewhat poor. For example, BCP mutations may be found in individuals with detectable HBeAg. Conversely, some individuals have anti-HBe in serum but wild type BCP sequences.

Incidence of HCC during 36 months follow-up

Sixty-one cases of HCC were diagnosed in the study cohort up to 36 months follow-up (Table 4, Figs 1 and 3); 54 are in the mutant group and only 7 are in the wild type group, giving HCC rates of 1463.4 vs. 235.14/100,000 person-years. The difference is significant ($p<0.001$), showing a strong association between the mutations and the development of HCC (rate ratio [RR]=6.23).

With further stratification according to gender, 42 male cases are in the mutant group and 3 in the wild type group, with rates of 2067.9 and 179.1 /100,000 person-years, respectively. Again, the difference is significant ($p<0.001$), showing a very strong association between the mutations and the development of HCC (RR=11.54). Specifically, 93.3% (42/45) of the male cases in the cohort are infected with the mutant type, suggesting that BCP mutations (A₁₇₆₂T, G₁₇₆₄A) may be a good biomarker to identify a subset of male HBsAg carriers at extremely high risk of HCC. Fewer cases of HCC have been diagnosed in the female study subjects, 12 mutant and 4 wild type ($P>0.1$) and further statistical analysis is not appropriate.

The significance of the mutations for females should become clear with a longer period of follow-up. Three additional cases of HCC occurred among the 740 individuals excluded from the analysis, a male and a female whose BCP sequences could not be determined and a female with a deletion (8 nt) in the region of interest.

In the HBsAg negative control group, only one (male) developed HCC, confirming that persistent HBV infection is the major cause of HCC in this region. These data confirm our hypothesis that the mutations may be used as a biomarker to identify males in Guangxi, and perhaps elsewhere in Asia, at high risk of developing HCC.

Risk factors of HCC

Male sex, older age, and BCP double mutations at baseline were all significantly associated with an increased rate of development of HCC in univariable analyses (left-hand panel, Table 5). There was also an apparently protective effect of baseline HBeAg, although this association did not reach statistical significance ($p=0.06$). Neither anti-HBe nor abnormal ALT levels at baseline were associated with the development of HCC. In a multivariate analysis, male sex, older age and BCP double mutations continued to be significantly associated with an increased development of HCC (middle panel, Table 5). Similar results were obtained when analyses were restricted to males only (right-hand panel, Table 5), with BCP double mutations and increasing age being independently associated with the development of HCC whereas no associations were found with HBeAg, anti-HBe or abnormal ALT concentrations.

Association with HCC of other mutations in the core promoter and enhancer II

Mutations at nucleotide (nt) 1653 from C to T and nt 1753 from T to A, G or C (T to V) have been reported to be associated with severe chronic hepatitis B and the development of cirrhosis and HCC (13). In order to determine the prevalence of these mutations in our study cohort and whether they also may be candidate biomarkers for HCC, we amplified and sequenced the X region of HBV from the study subjects with HCC and controls. The total prevalence of the C₁₆₅₃T mutation from cases and controls is 20% (12/60) and that of T₁₇₅₃V is 28.4% (29/102) (Table 6). The differences between HCC cases and controls are not significant at either position. Almost all mutations at nt 1653 and nt 1753 are co-incident with the A₁₇₆₂T, G₁₇₆₄A mutations. Whilst it was intended that all cases would be individually matched to a control (by sex, age, neighborhood and BCP sequence), this was not possible for all cases (two for nt 1653 and one for nt 1753). Thus, the main analysis (Chi-squared test) incorporated all available cases and controls but did not take account of the matching. However, a sensitivity analysis, which included only those cases and controls that were matched successfully (28 for nt 1653 and 33 for nt 1753) and utilized McNemar's test to take account of the matching, reached similar conclusions.

DISCUSSION

To our knowledge, this prospective study is the first designed to determine whether the incidence of HCC in subjects infected with HBV with BCP double mutations is significantly higher than that in subjects infected with HBV with wild type BCP and to clarify whether these double mutations constitute a valuable biomarker of an extremely high risk of developing HCC. After 36 months' follow-up, our data show that, in males aged >30, these mutations are an aetiological risk factor of HCC. Of note, 93.3% of male cases were infected with mutant HBV, suggesting that the A₁₇₆₂T, G₁₇₆₄A mutations are a valuable biomarker to identify a subset of male HBsAg carriers aged >30 at extremely high risk of hepatocellular carcinoma. The major weakness of this study is that the role of viral loads which have been reported to be associated with the development of HCC (10, 14, 15) has

not been studied in detail because of the high cost of quantitation, although measurement of viral loads in a representative sample of individuals from this cohort showed that the BCP double mutations are not associated with higher viral loads (in fact, HBeAg-positive individuals with BCP double mutations have significantly lower viral loads than those with wild type HBV; data not shown). Another weakness is that, with this limited period follow-up, it is not clear whether the BCP double mutations are associated with the development of HCC in females, because only a small number of HCC cases has been diagnosed because of the lower incidence of HCC in females (HCC is approximately four times more common in males than females).

Although HBV genotypes have been reported to be of clinical relevance (9), their role in the development of HCC remains controversial (10-14, 27). Furthermore, our random genotyping of this cohort shows that the distribution of genotypes in subjects without HCC is similar to that in HCC (data not shown). Therefore we did not determine the HBV genotype for all study subjects. The incidence of HCC in HBV cirrhotic patients has been found to be greater than in non-cirrhotic patients (28). In this study, 67 individuals were known to have cirrhosis and 39 of the 61 HCC cases occurred in individuals with cirrhosis (data not shown). However, BCP double mutations, *per se*, are an independent risk factor for the development of liver cirrhosis (18, 29) and cirrhosis was not included in this analysis.

Since Chu et al. (19) and we (17) reported that BCP double mutations are associated with the development of HCC, there have been many studies that confirm these findings (14, 20, 21). It has been suggested that the double mutations may serve as a useful molecular marker for predicting the clinical outcomes of patients with chronic hepatitis B (21) and might contribute to the gender difference in the progression of liver disease in HBeAg-negative chronic hepatitis B (30). A retrospective nested case-control study showed that the BCP double mutations were statistically significantly associated with HCC risk, even after adjusting for ALT levels, anti-HBe, HBV genotype, HBV viral load, and other sequence variants (31). However, these are all cross-sectional or case-control studies and cannot prove the etiological relationship between BCP mutations and the development of HCC, or determine the incidence of HCC. One retrospective longitudinal analysis determined that the patients had these mutations prior to diagnosis of HCC (32) and confirms that the BCP mutations play an etiological role in the development of HCC. However, this is a small retrospective study (120 study subjects); sampling bias could not be avoided and it was not possible to determine the incidence of HCC in the BCP mutant and wild type groups. Our prospective study overcomes all of these problems.

Other studies have suggested that the core promoter mutations C₁₆₅₃T and T₁₇₅₃V also are associated with the occurrence of severe hepatitis B and development of HCC, although their prevalence is lower than that of the A₁₇₆₂T, G₁₇₆₄A mutations (13, 33-35). In this study cohort, the C₁₆₅₃T and T₁₇₅₃V mutations were detected less frequently (from 20%-28.4%) than the A₁₇₆₂T, G₁₇₆₄A mutations and the differences between HCC cases and controls are not significant for either. Furthermore, the C₁₆₅₃T and T₁₇₅₃V mutations occur almost invariably on the background of the A₁₇₆₂T, G₁₇₆₄A mutations. Therefore, we conclude that the A₁₇₆₂T, G₁₇₆₄A mutations are a valuable biomarker to identify those male HBsAg carriers who should be monitored regularly for signs of tumour.

It might seem that the value of this biomarker is limited, given that the mutations were detected in around half of the HBV DNA-positive individuals in our cohort. However, we believe that this extremely high prevalence of the mutations is attributable to our conducting this prospective study in an area amongst the highest in the world in terms of annual incidence of HCC. Our earlier studies (17, 18) reported a prevalence of the mutations in around one quarter of HBV DNA-positive individuals elsewhere in Guangxi and studies in

neighboring provinces show a prevalence ranging from 10% (Guangdong (36)) to 30% (Hunan (37)). Therefore, we consider it worthwhile to screen HBsAg carriers for the mutations in areas where there is a high annual incidence of HCC. Currently, in Guangxi, China, HBsAg positive males aged >30 are considered as higher risk population of HCC (22-24) and recommended to undergo regular surveillance for HCC. Considering the high prevalence of HBsAg and the limited public health resources in Guangxi and in other highly endemic areas, it is recommended to narrow down further the number of subjects requiring surveillance for HCC using the biomarker of BCP double mutations, despite that this will result in failure to detect around 7% of HCC cases, which occur in the male population infected with HBV with wild type BCP sequences.

We found at baseline that the prevalence of HBV with BCP double mutations increases with age and differs significantly between age groups 30-34 and 35-39, and 35-39 and 40-44. In future, we will repeat the assays on samples collected longitudinally to determine the rate at which this shift occurs and whether wild type HBV re-emerges in individuals in whom the variant viruses were selected previously.

High viral loads ($\geq 10^4$ copies/ml) have been reported to be associated with the development of HCC (10, 14, 15) and also are of importance in this study cohort. As described in the paragraph covering grouping of study subjects, 1402 HBsAg positive individuals were HBV DNA negative with the PCR conducted originally in China. Subsequently, we measured the viral loads of 34 random male subjects from the DNA 'initially DNA negative' group and 98 random male subjects from the DNA positive group. Serum viral DNA concentrations were $\geq 10^4$ copies/ml in 17.6% (6/34) of the DNA 'negative' group and 89.8% (88/98) of the DNA positive group (data not shown). After 36 months follow-up, only 10 HCC cases were found in these "initially DNA negative" male subjects (1.39%, 10/719) compared to 36 cases in DNA positive group (3.92%, 36/919). We also measured the baseline viral loads of 20 male HCC cases from which sufficient serum remained available; 85% (17/20) are $\geq 10^4$ copies/ml (data not shown). In the future, we shall assess further the role of viral loads in the development of HCC in this cohort, and determine whether this factor acts independently of, or synergistically with, the BCP mutations and whether viral loads can be used as an additional biomarker to narrow down further the population at risk (i.e. those identified with BCP double mutations). Meanwhile, we will continue to follow up our cohort to determine whether the BCP mutations have the same value as a biomarker in females.

STUDY HIGHLIGHTS

1) What Is Current Knowledge

- Surgery remains the only curative modality for hepatocellular carcinoma (HCC)
- Surveillance can detect small HCC for resection but at huge cost in health resources
- It is difficult to narrow down the population for surveillance of HCC

2) What Is New Here

- Hepatitis B virus (HBV) A₁₇₆₂T, G₁₇₆₄A mutations are a valuable biomarker to identify males at extremely high risk of HCC
- The number of males for surveillance for HCC can be narrowed down according to the presence of HBV A₁₇₆₂T, G₁₇₆₄A mutations
- HBV A₁₇₆₂T, G₁₇₆₄A mutations are an etiological risk factor of HCC

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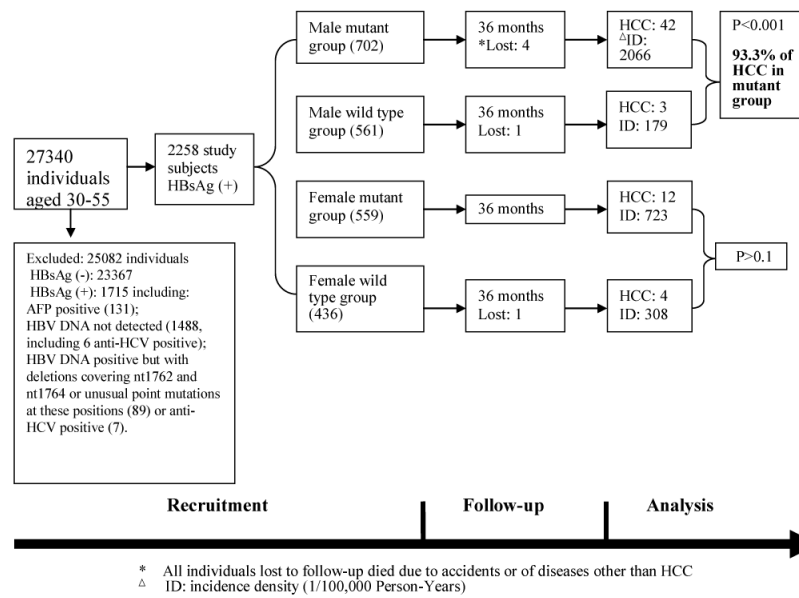


Figure 1. Flow diagram of the progress through the phases of the prospective study (recruitment, follow-up and data analysis).

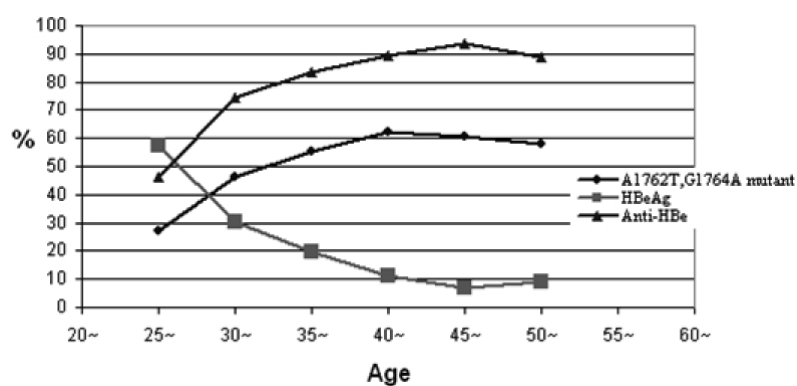


Figure 2. prevalence of the A1762T, G1764A mutations, HBeAg and anti-HBe with age

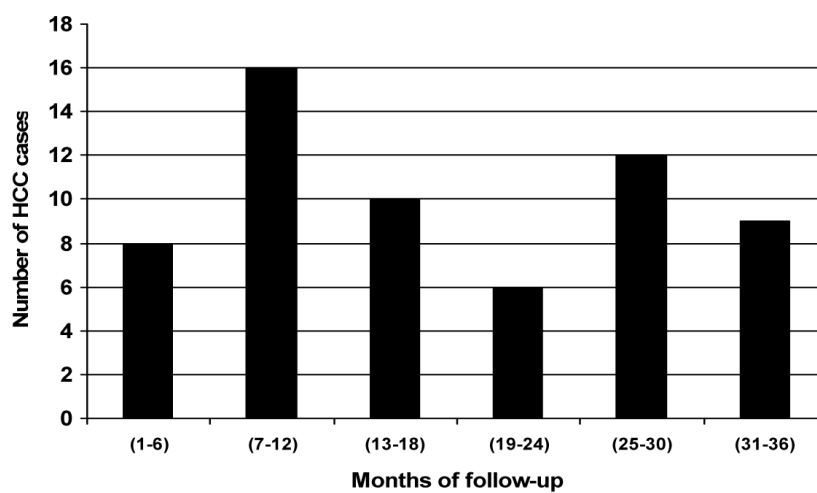


Figure 3. HCC cases diagnosed during 36 months of follow-up.

Table 1

Frequency of HBsAg according to gender and age

Age	No. Subjects			HBsAg (+)			HBsAg (+) Rate (%)		
	Male	Female	Total	Male	Female	Total	Male	Female	Total
25~	126	131	257	30	19	49	23.8	14.5	19.1
30~	3465	3505	6970	726	500	1226	21.0	14.3	17.6
35~	2999	3121	6120	531	411	942	17.7	13.2	15.4
40~	2630	2953	5583	423	375	798	16.1	12.7	14.3
45~	1711	2164	3875	225	251	476	13.2	11.6	12.3
50~	1961	2574	4535	214	268	482	10.9	10.4	10.6
Total	12892	14448	27340	2149	1824	3973	16.7 (16.0-17.3)	12.6 (12.1-13.2)	14.5 (14.1-14.9)

Males vs. Females (HBsAg positivity rate): $X^2=89.737$, $P<0.001$



Table 2
HBV DNA and prevalence of HBV BCP double mutations according to gender

Gender	N	HBV DNA(+)	A ₁₇₆₂ T, G ₁₇₆₄ A mutations	Unusual mutations at A ₁₇₆₂ , G ₁₇₆₄	Prevalence of A ₁₇₆₂ T, G ₁₇₆₄ A mutations
Male	1638	1310 [†]	704	42 [*]	53.7 (51.0-56.4)
Female	1360	1044 [†]	560	47 ^{**}	53.6 (50.6-56.6)
Total	2998	2354[†]	1264	89	53.7 (51.7-55.7)

Males vs. Females (Prevalence of A₁₇₆₂T, G₁₇₆₄A double mutations): $\chi^2 = <0.0001$, $P > 0.99$

[†] individuals for whom HBV DNA was detected and BCP sequences determined, including 7 anti-HCV positive who were excluded from the analysis of HCC incidence

^{*} Includes 3 with deletions in this region

^{**} Includes 5 with deletions in this region

Table 3
Prevalence of HBV BCP double mutations according to age

Age	N	HBV DNA (+)	A ₁₇₆₂ T, G ₁₇₆₄ A mutations	A ₁₇₆₂ T, G ₁₇₆₄ A mutations (%)
25-29	40	29	8	27.6
30-34	970	742	336	45.3
35-39	701	554	306	55.2
40-44	595	480	292	60.8
45-49	349	280	169	60.4
50-54	343	269	153	56.9
Total	2998	2354	1264	53.7 (51.7-55.7)

Group 30-34 vs. 35-39: $\chi^2=12.171$, $P<0.001$

Group 35-39 vs. 40-44: $\chi^2=3.080$, $P=0.08$

Table 4
HCC incidence in mutant and wild type groups after 36 months follow-up

Groups	N	Person-years	HCC	Incidence Rate (/100,000 years)	95% CI	Rate ratio	95% CI	p-value
Overall	2258	6669	61	914.7	685.1, 1144.2			
Mutant	1261	3690	54	1463.4	1073.1, 1853.7	6.23	2.83, 13.68	0.0001
Wild type	997	2978	7	235.1	94.5, 484.2			
Males								
Mutant	702	2031	42	2067.9	1442.5, 2693.3	11.54	3.58, 37.24	0.0001
Wild type	561	1675	3	179.1	37.0, 523.4			
Females								
Mutant	559	1659	12	723.2	373.8, 1263.4	2.36	0.76, 7.31	0.14
Wild type	436	1304	4	306.7	83.6, 785.3			
HBsAg (-) controls	2460	5985	1	16.7	0.4, 93.1			

Table 5
Analysis of factors associated with the development of HCC using Poisson regression models

Factor	Unadjusted			Adjusted*			Adjusted (males only)		
	RR	(95% CI)	p-value	RR	(95% CI)	p-value	RR	(95% CI)	p-value
Male sex	2.25	(1.27-3.98)	0.005	2.67	(1.50-4.75)	0.0004	n/a		
Age group									
30-34	1		0.0007	1		0.0009	1		0.003
35-39	3.00	(1.30-6.96)		2.74	(1.18-6.35)		3.78	(1.36-10.48)	
40-44	1.99	(0.78-5.03)		1.76	(0.69-4.47)		2.71	(0.91-8.09)	
45-49	4.25	(1.74-10.39)		3.88	(1.58-9.53)		4.57	(1.49-13.99)	
50-54	5.16	(2.17-12.31)		5.36	(2.23-12.86)		7.04	(2.36-21.00)	
BCP double mutations	6.23	(2.83-13.68)	0.0001	5.78	(2.63-12.73)	0.0001	10.73	(3.32, 34.66)	0.0001
HBeAg	0.42	(0.17-1.05)	0.06						
Anti-HBe	0.92	(0.46-1.82)	0.81						
Abnormal ALT	1.58	(0.82-3.05)	0.17						

* After adjustment for sex, age and BCP double mutations, HBeAg was no longer associated with the rate of HCC and so was not included in the multivariable model.



Table 6
Occurrence of nucleotide 1653 and 1753 mutations in individuals with HCC and controls

Groups	C ₁₆₅₃ T			T ₁₇₅₃ V		
	N	Mutation	Incidence (%)	N	Mutation	Incidence (%)
HCC	30	8 (7)*	26.7	34	13 (13)*	38.2
Control	30	4 (4)	13.3	68	16 (13)	23.5
Total	60	12 (11)	20.0	102	29 (26)	28.4

$\chi^2_{1653}=0.938$ P=0.33

$\chi^2_{1753}=1.741$ P=0.19

* Numbers in parentheses indicate the coexistence of the C₁₆₅₃T and T₁₇₅₃V mutations with the A₁₇₆₂T, G₁₇₆₄A mutations.